Advanced Microcontroller Bus Architecture – Advanced Peripheral Bus

An Undergraduate Project Report submitted to Manipal Academy of Higher Education in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY
In

Electrical & Electronics Engineering

Submitted by

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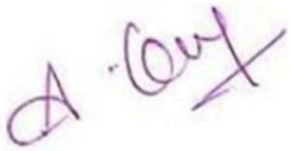
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CERTIFICATE

This is to certify that the project titled **Advanced Microcontroller Bus Architecture – Advanced Peripheral Bus** is a record of the bonafide work done by **Tarini Kavuturu** (*Reg. No. 220906496*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in **ELECTRICAL AND ELECTRONICS ENGINEERING** of Manipal Institute of Technology, Manipal, Karnataka, (A Constituent unit of Manipal Academy of Higher Education), during the academic year 2025-2026.



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SUB: - COMPLETATION OF INTERNSHIP PROGRAM

This is to certify that **Ms. Kavuturu Tarini** has completed her Internship Program with **M/s Trident Techlabs Limited,** from **02nd February, 2026 to 02nd June, 2026** under the mentorship of **Mr. Jiyo Mathew.**

During Internship tenure she has learned about VLSI process. She was based at our **Bangalore office.**

In this span of training her candidature was found sincere.

We wish her best of luck for the future endeavor.

Thanking you

Your sinercely

for Trident Techlabs Limited

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ABSTRACT

The report documents the internship carried out at Trident Techlabs Ltd., Bangalore. The main objective of the internship was to design and verify AMBA APB protocol at an RTL level. With the changing times, humans have had to adapt to faster and more efficient ways of doing everyday tasks. An industry that flourished in such a time is digital electronics and it has seen rapid growth ever since. To facilitate such advancement, a need emerged for efficient communication between the multiple components on an SoC. ARM produced a family of protocols, AMBA to resolve this issue. AMBA stands for Advanced Microcontroller Bus Architecture and it is a family of protocols that enables fast and efficient communication between the processors, memory units, the peripherals, etc. But each of the components do not have the same power and complexity requirements. AMBA has introduced multiple protocols to address the varying needs of these components. Among these, the AMBA APB caters to the low-power, low-bandwidth communication between the processor and peripherals such as UART, GPIO, timers, etc.

The internship began with the study of Verilog HDL and implementation of some basic digital circuits. This included writing RTL codes for multiple systems such as counters, flip-flops, etc. The written RTL codes were verified using the tools provided in the Siemens EDA toolkit. Questa Lint was used for static analysis, Questa Sim for dynamic verification through simulation, HDL Designer for visualizing the structure of code and finally Questa Inspect to carry out beginner level formal verification. The internship then progressed to the study of AMBA APB and its RTL level implementation. The RTL implementation of the APB protocol consists of 3 modules – the Operating States module, the Dataflow module and the Top module.

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LIST OF ABBREVIATIONS

AMBA	Advanced Microcontroller Bus Architecture
APB	Advanced Peripheral Bus
AHB	Advanced High-performance Bus
AXI	Advanced eXtensible Interface
RTL	Register Transfer Level
FSM	Finite State Machine
EDA	Electronic Design Automation
SoC	System on Chip
HDL	Hardware Description Language
PCLK	APB Clock Signal
PRESETn	APB Reset Signal
PADDR	APB Address Signal
PSELx	APB Peripheral Select Signal
PENABLE	APB Enable Signal
PWRITE	APB Write Direction Signal
PWDATA	APB Write Data Signal
PRDATA	APB Read Data Signal
PREADY	APB Ready Signal
PSLVERR	APB Transfer Error Signal
RME	Realm Management Extension
UART	Universal Asynchronous Receiver Transmitter
GPIO	General Purpose Input Output
I2C	Inter Integrated Circuit
SPI	Serial Peripheral Interface
FPGA	Field Programmable Gate Array

CHAPTER 1
INTRODUCTION

The rapid advancement in digital electronics has led to a fat growing demand for new efficient, low-power on-chip communication in System-on-Chip (SoC) designs. SoCs integrate multiple components such as processors, memory units and peripherals. This requires standardized communication interfaces for reliable data transfer. The widespread adoption of IoT, automotive electronics, consumer electronic devices, automation of industrial processes has further increased the need of sophisticated and complex SoCs, creating a growing need for communication protocols to satisfy the multitude of requirements.

To address these challenges, ARM's Advanced Microcontroller Bus Architecture (AMBA) provides a family of protocols to address varying performance requirements within an SoC. These include Advanced Extensible Interface (AXI), Advanced High-Performance Bus (AHB) and Advanced Peripheral Bus (APB). Among these, APB is specifically designed for low-bandwidth, low-power communication with peripherals such as UART, GPIO, I2C, SPI, timers and interrupt controllers.

APB is a non-pipelined, synchronous protocol in which every transfer takes a minimum of 2 clock cycles. This simplicity results in minimal power consumption and reduced interface complexity, making it a preferred choice for peripheral register access. The protocol operates through 3 well defined operating states – IDLE, SETUP and ACCESS – and supports both read and write transfers, with and without wait states.

The internship was carried out at Trident Techlabs Ltd., Bangalore. The primary objective was to learn the basics of Verilog and the usage of verification tools and use them to design and verify AMBA APB protocol at RTL level.

The motivation behind this project is to bridge the gap between theoretical learning and practical implementation. Since AMBA APB is a widely used protocol, implementing the protocol provides valuable insight into the functioning of such protocol and also improves proficiency in RTL coding, verification workflows, debugging techniques and the effective use of professional Siemens EDA tools.

The project is highly relevant in the VLSI industry as communication between the peripherals and the processor directly influences the overall efficiency and reliability of an SoC.

During the course of the internship, industry level standards were employed and verification of the programs were thorough. The work combines RTL design, lint analysis, simulation and formal verification using Siemens EDA tools, following the industry process of verification flow employed in professional semiconductor development.

The project is entirely simulation-based and does not involve use of any dangerous materials or equipment. Therefore, there is no direct implication on human health or safety. However, even a small error in the code could lead to catastrophic events. So, care was taken to verify all the codes with utmost precision, by employing a high level of quality checks using the Siemens EDA tools. From an environmental perspective, simulation before hardware implementation minimizes material consumption and electronic waste.

The project was carried out adhering to fundamental principles of engineering, including honesty, integrity, professionalism and accountability. The results and conclusion presented in the report accurately reflect the work that was carried out without alteration or misinterpretation. Licensed Siemens EDA tools were used in accordance with institutional policies.

CHAPTER 2
LITERATURE REVIEW AND OBJECTIVES

The literature review provides a detailed description of the existing concepts, methodologies and standards that are related to AMBA APB and verification in digital systems. It includes the evolution of AMBA APB and the significance of APB in modern day SoCs.

The AMBA APB protocol has been part of the ARM ecosystem since its first release and has undergone several revisions to meet growing design requirements. As stated in the ARM IHI 0024E and the other related literature, they are as follows.[1]

2.1 Evolution of AMBA APB [8]

The APB specification has evolved through multiple issues:

- APB2: The original APB defined the basic two-phase handshake with PCLK, PRESETn, PADDR, PWRITE, PSEL, PENABLE, PWDATA and PRDATA. [1]
- APB3: Introduced PREADY, allowing the peripheral to insert wait states and PSLVERR for error reporting. [1]
- APB4: Added PPROT for transaction-level access protection and PSTRB for sparse write data transfers. [1]
- APB5: Introduced PWAKEUP for power management, user-defined signals (PAUSER, PWUSER, PRUSER, PBUSER), interface parity protection and optional Realm Management Extension (RME) support via PNSE. [1]

2.2 APB Signal Interface

The APB interface consists of signals driven by either the Requester (APB master) or the Completer (peripheral). Key signals include: [1]

- PCLK: Clock signal; all APB signals are sampled at the rising edge.
- PRESETn: Active-low reset signal.
- PADDR: Address bus, up to 32 bits wide.

- PSELx: Select signal, one per peripheral, asserted by the Requester to initiate a transfer.
- PENABLE: Enable signal, asserted in the second cycle of every transfer.
- PWRITE: Direction signal; HIGH for write, LOW for read.
- PWDATA / PRDATA: Write and read data buses (8, 16, or 32 bits).
- PREADY: Handshake signal from the Completer; held LOW to insert wait states.
- PSLVERR: Error signal from the Completer, valid only when PSEL, PENABLE and PREADY are all HIGH.

2.3 Operating States

The APB state machine defines three operating states as per the specification:

- IDLE: Default state when no transfer is pending. PSELx is deasserted.5
- SETUP: Entered when PSELx is asserted. Address, control and data signals are driven and stabilised. The interface remains in SETUP for exactly one clock cycle.
- ACCESS: Entered on the next rising edge after SETUP, with PENABLE asserted. The transfer completes when PREADY is driven HIGH by the Completer. Signals such as PADDR, PWRITE, PWDATA and PPROT must not change during this state.

2.4 Read and Write Transfers

The APB protocol supports both read and write data transfers through a simple two-phase operation.

- Write Transfer: During a write operation, the Requester asserts PSEL, drives the target address on PADDR, places the write data on PWDATA and sets PWRITE HIGH. The transfer completes in the ACCESS state when PENABLE and PREADY are asserted.
- Read Transfer: During a read operation, the Requester asserts PSEL and provides the target address while keeping PWRITE LOW. The Completer places the requested data on PRDATA and the transfer completes when PREADY is asserted during the ACCESS state.

2.5 Verification Methodologies [9]

Modern digital design verification employs multiple techniques to ensure design correctness:

- Linting: Linting is a static code analysis technique which is used to examine the HDL code without executing it. [3]
- Simulation: The functional correctness of a code is verified using testbenches. Waveforms are observed to analyse the signal timings and any logical errors. [2]
- Coverage Analysis: Metrics such as statement coverage, branch coverage, FSM coverage, etc are used to assess the degree of completion to which the testbench tests the code.[2]
- Formal Verification: The code is converted into a mathematical problem which is applied to prove or disprove design properties. It is useful for detecting corner cases such as deadlocks, unreachable statesand protocol violations. [4]

The Siemens EDA Questa tool suite — comprising of Questa Lint, Questa Sim and Questa Inspect — supports all the above techniques and is widely used in the semiconductor industry.

OBJECTIVES

The objectives of this internship project are:

1. To design and implement the AMBA APB protocol at the RTL level, beginning with the Operating States FSM module.
2. To verify each implemented module through static analysis, simulation and basic formal property checking.
3. To establish a structured understanding of the industry verification flow as applied to a real communication protocol.

CHAPTER 3
METHODOLOGY

This chapter presents the methodology followed during the duration of the internship. The work involved learning RTL design fundamentals, implementing AMBA APB at an RTL level, performing simulations and verifying the quality of the code using Siemens EDA tools. The methodology implemented assured systematic approach from basics of RTL design to verification.

3.1 Nature of the Project

The project work is primarily focusses on digital hardware design and verification through RTL implementation using Verilog HDL. The project includes design development, functional simulation, static analysis, design visualization and introductory formal verification of AMBA APB protocol. The project is carried out using Questa EDA tools, without any hardware implementation.

3.2 Detailed Methodology

3.2.1 RTL Design Fundamentals

The internship began with the study of Verilog HDL and implementation of some basic digital circuits including adders, multiplexers, decoders, flip-flops, counters and FSM. [7] These RTL codes were written and then simulated on Questa Sim. This exercise ensured that correct coding practices were established and there was a hands-on understanding of the systems.

3.2.2 Static Analysis — Questa Lint

The previously written RTL codes were statically analysed using Questa Lint. This helped in identifying issues such as – syntax violations, incomplete statements, missing default cases, inferred latches, unused signals, potential dead code, etc – before simulation. Lint results are reviewed and RTL code is corrected to improve the code quality.

3.2.3 Simulation and Debugging — Questa Sim

Testbenches were written and applied to the RTL code for dynamic verification of the design. Questa Sim was used to generate waveforms which were analysed to find inconsistencies in the design. Code coverage reports were generated for all the designs to assess the thoroughness of the execution of the code. Higher code coverage meant improved trust in the simulation verification

carried out.

3.2.4 Design Visualization — HDL Designer

HDL Designer is used to visualize the RTL structure of the modules. This includes module hierarchy diagrams, block diagrams and FSM state diagrams. This provided a graphical view of how the design was structured and the interconnection of all the components in the RTL code.

3.2.5 AMBA APB Protocol Study

The complete AMBA APB Protocol Specification (ARM IHI 0024E) was studied in detail. This covered signal descriptions, timing diagrams for read and write transfers (with and without wait states), error response behaviour and the three-state operating model. This study formed the basis for the RTL implementation of the APB model.

3.2.6 APB Operating States Module — Implementation

The Operating States module was implemented as a three-state FSM in Verilog. The module handles transitions between IDLE, SETUP and ACCESS states based on the input signals PSELx, PENABLE and PREADY . The implementation was verified using Questa Sim and Questa Lint.

3.2.7 APB Dataflow Module — Implementation

The Dataflow Module was implemented to carry out read and write transfers between the master and slave. It processes the data transfers based on the input signal and generates the

appropriate output signals. The Data Transfer module receives the APB control signals PSEL, PENABLE, PWRITE, along with PADDR and PWDATA as inputs to perform read and write operations. Based on these inputs, it generates the output signals PRDATA and PREADY, ensuring correct data transfer during APB transactions. The Top Module integrates the Operating Systems Module and the Dataflow Module to form a complete model of the APB protocol.

3.2.8 Introductory Formal Verification — Questa Inspect

Questa Inspect is used as a beginner platform to explore formal verification concepts. It provides a beginner friendly approach with automatic checks done to find any dead code or forbidden input sequences that may lead to crashing of the entire system.

3.3 Assumptions

The project assumes a single APB master communicating with a single slave. An internal memory block is used for data storage during simulation and the design follows APB protocol without considering the advanced features such as error response, multiple slaves, etc.

3.4 Module Specifications

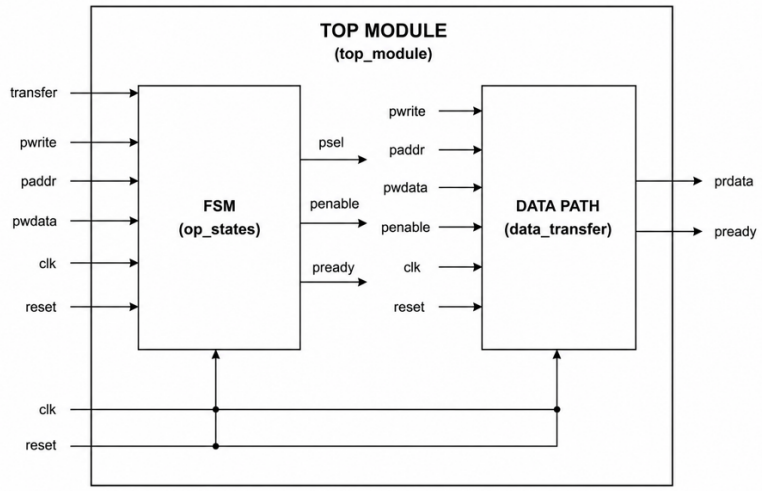


Fig 1: Block Diagram of the Modules

The project consists of 3 modules – the Operating States module which manages the state transition, the Dataflow module which is responsible for performing the read and write transactions and the Top module which integrates both modules to implement complete APB communication. Each module is individually verified and tested as well as after integration.

3.5 Hardware / Software Requirements

The project utilizes Verilog HDL for RTL implementation and Siemens Questa EDA tools. They include Questa Sim for functional simulation and code coverage, Questa Lint for static RTL analysis, HDL Designer for design visualization and Questa Inspect for introductory formal verification. These tools were selected because they provide an integrated environment for designing, analysing, simulating and verifying digital hardware designs efficiently.

CHAPTER 4
RESULT ANALYSIS

This chapter details the results obtained during the implementation and verification of the APB protocol. The RTL code was verified using the Siemens EDA tools – Questa Lint for statical analysis, Questa Sim for dynamic verification, HDL Designer for visualizing the hierarchy of the code and finally Questa Inspect to perform beginner friendly formal verification. The objective of using these tools is to validate the correctness of code and ensure compliance with industry standards.

4.1 Questa Lint Results

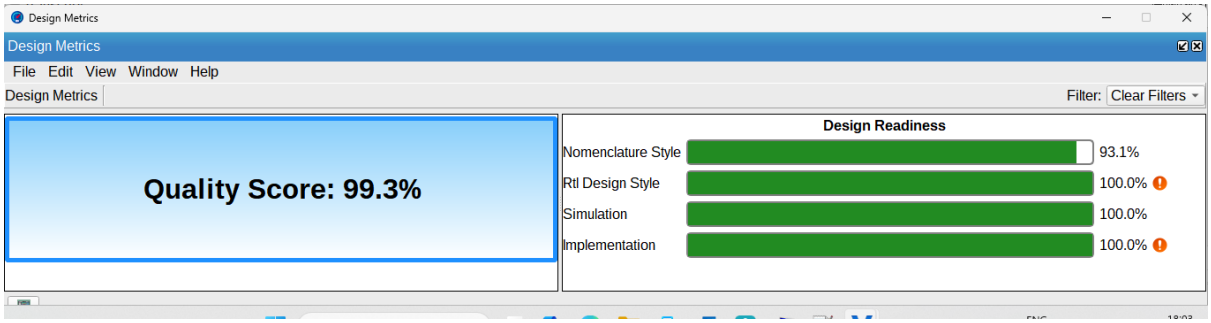


Fig 2: Questa Lint – Quality Score results

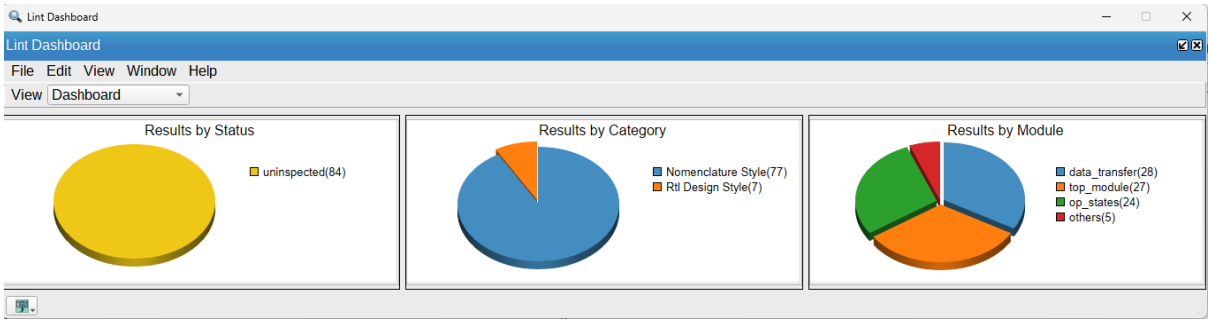


Fig 3: Questa Lint – Dashboard results

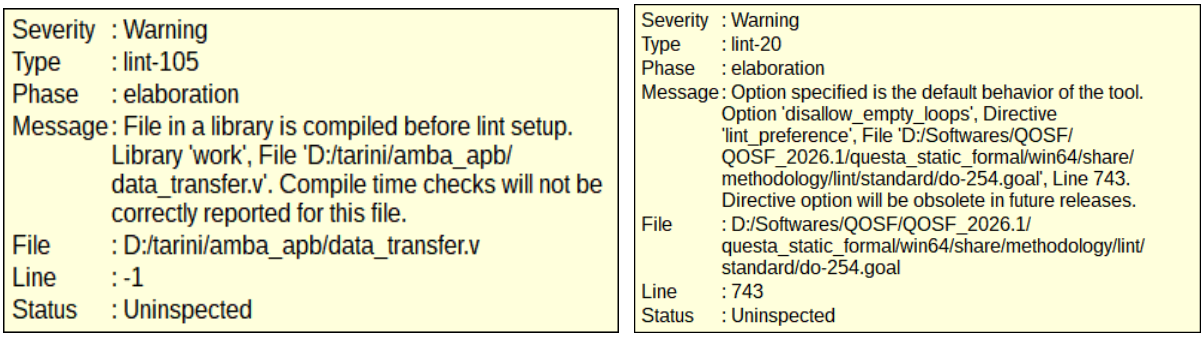


Fig 4: Quest Lint – Error Messages

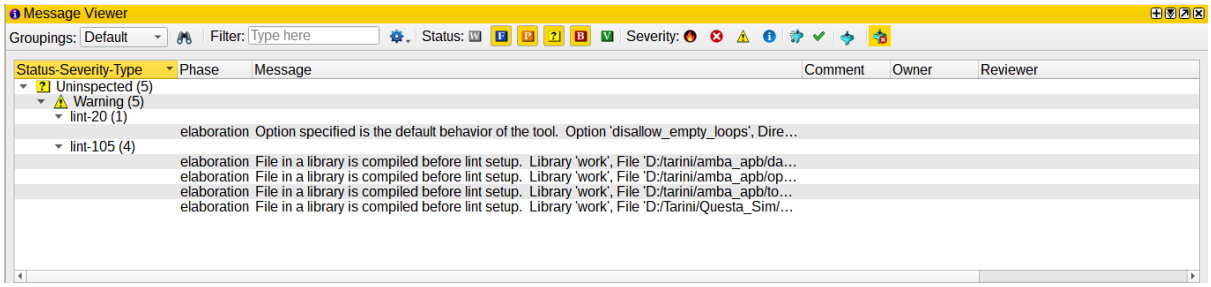


Fig 5: Quest Lint – Warning Messages

The Lint results show a Quality Score of 99.3%. This is considered a good score and proves the correctness of the code as well as proper coding processes being followed. The minor reduction in score is majorly caused because of the naming in the code slightly deviates from the industry standard nomenclature rules. The warning messages are also indicative that there was a problem in the launch of lint analysis but this however, does not affect the functionality of the code. The results indicate that the design follows the recommended coding guidelines with only minor issues identified during lint analysis.

4.2 Questa Sim Results

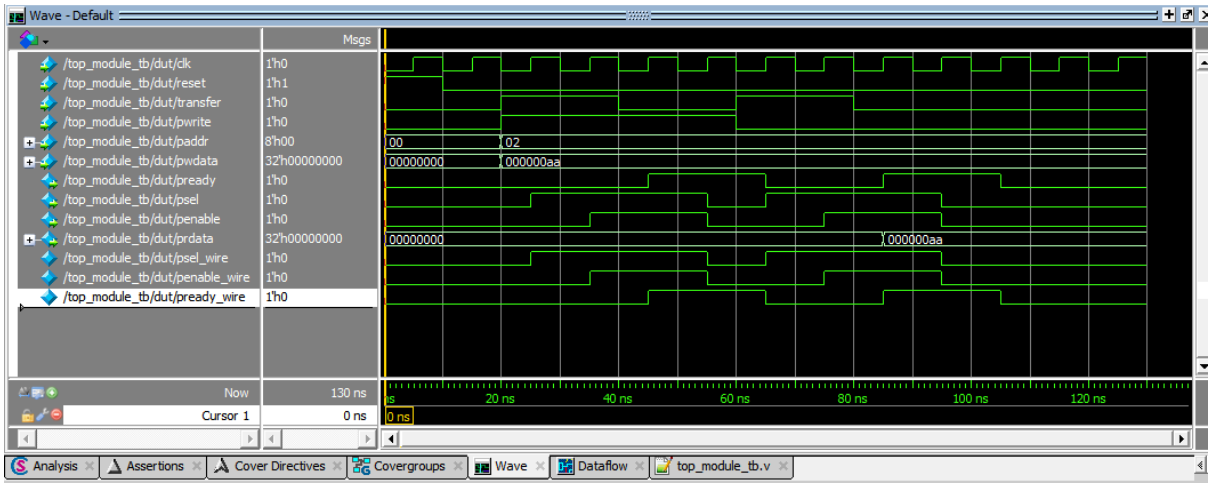


Fig 6: Questa Sim – Waveforms

The results are as expected. After reset, a transfer request is asserted and the FSM transitions from IDLE state to the SETUP state. During the SETUP phase, PSEL is asserted and a value is assigned to the PADDR. At the rising edge of the next clock cycle, PENABLE is asserted

while PSEL remains high. Once PREADY is asserted, the transfer takes place and the FSM returns to the IDLE state where PSEL and PENABLE are deasserted. This signifies that the system is ready for the next transaction. Both read and write transfers are successfully verified. [10]

4.3 Code Coverage Results on Questa Sim

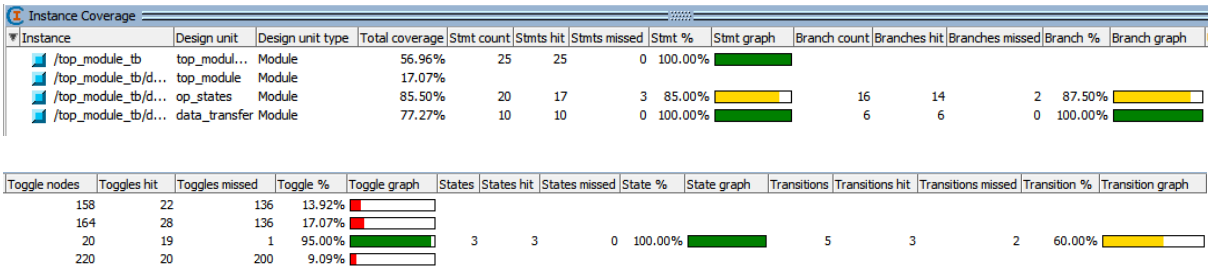


Fig 7: Questa Sim – Code Coverage results

According to the code coverage analysis, most of the statements and branches have been covered using the testbench given. All 3 states in the FSM have been hit and 60% of the transitions were tested. Lower toggle coverage indicates some signals have not been tested exhaustively by the specified test cases. [10]

4.4 HDL Designer Results

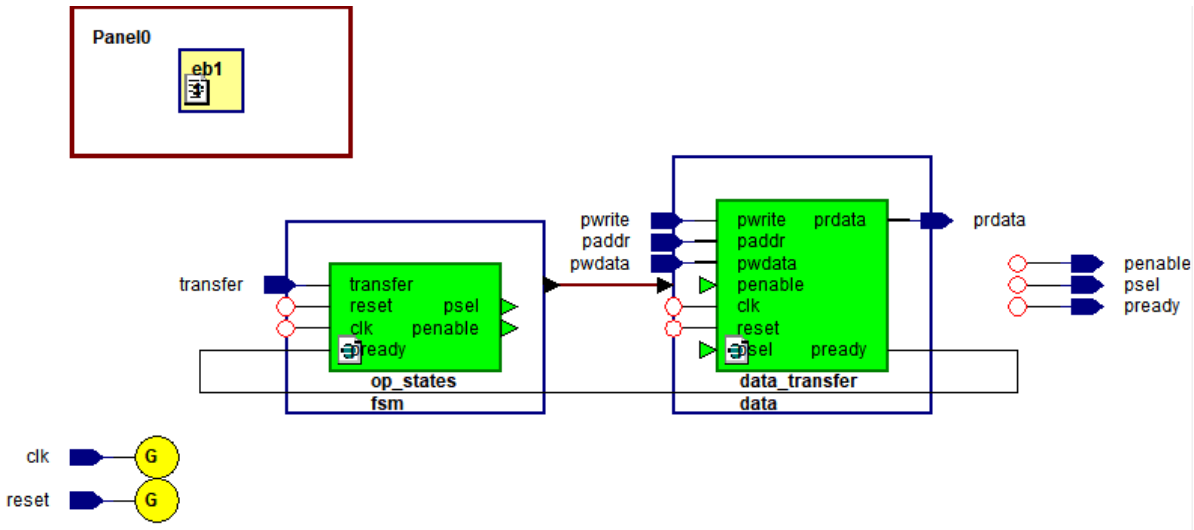


Fig 8: HDL Designer – Block View

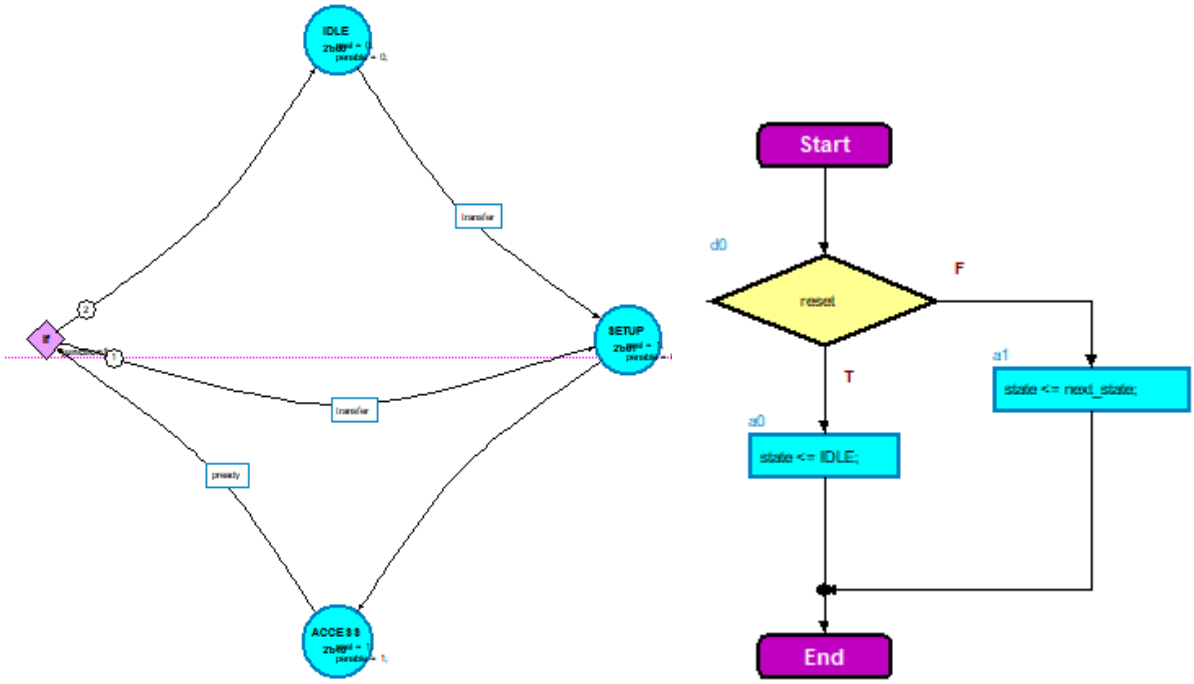


Fig 9: FSM and Flowchart View of Operating States Module

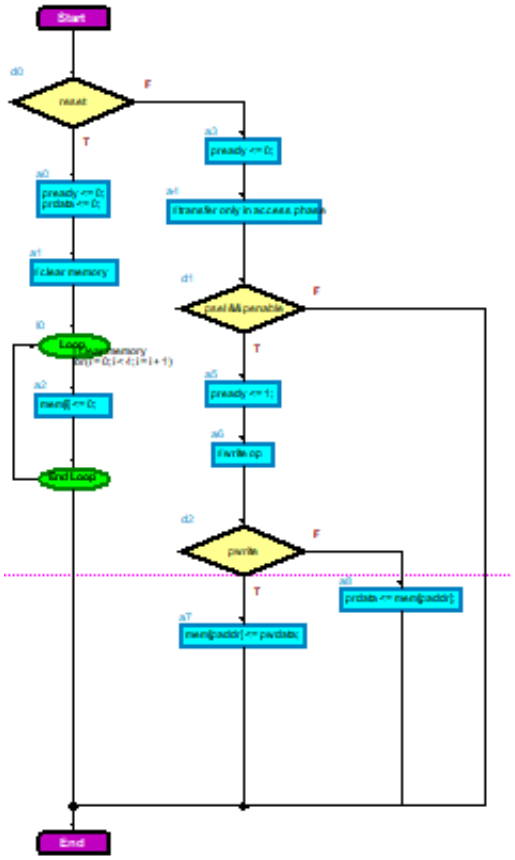


Fig 10: Flowchart of Dataflow Module

HDL Designer is used to understand the structure and hierarchy of the RTL code. The above figures show the structure, hierarchy and state transitions and the flow in both the modules with the help of flow diagrams and block diagrams.

4.5 Questa Inspect Results

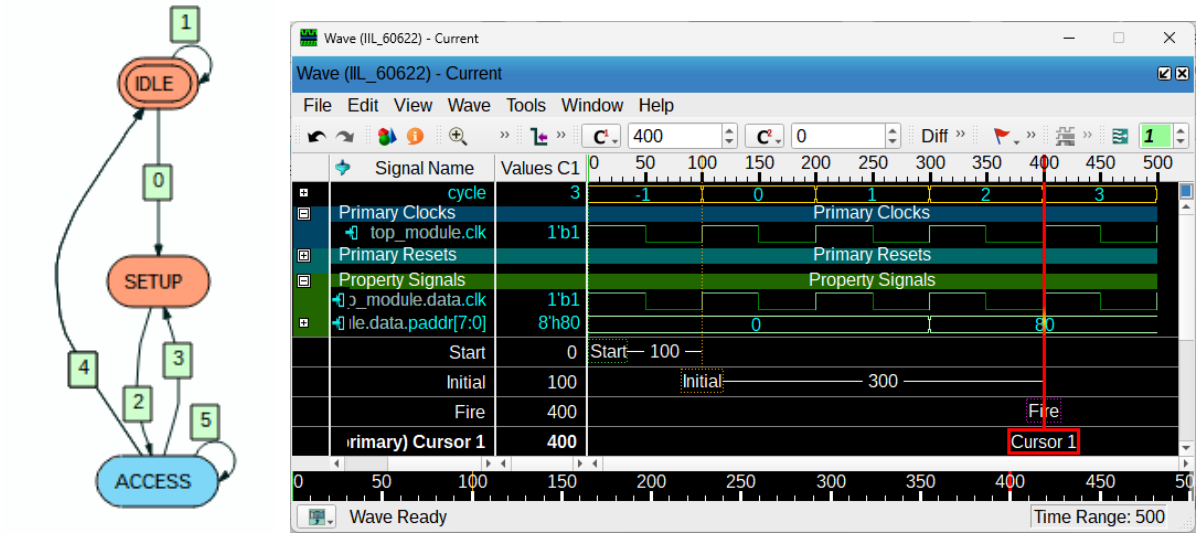


Fig 11: Questa Inspect Results

The figure above illustrates a waveform generated by Questa Inspect that shows the behaviour of the APB RTL design over time by displaying the changes in clock, address and verification property signals during simulation. The FSM graph represents the state transitions of the APB controller, confirming the correct sequence of IDLE, SETUP and ACCESS states. These prove the functional correctness of the code.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

The main objective of the internship was to achieve an RTL implementation of an established protocol, AMBA APB. The objectives and expectations set were met. A thorough understanding of the RTL design practices and the verification flow were achieved. The hands on implementation technique followed throughout the internship was helpful in developing proficiency in basic digital circuits, HDL coding as well as Siemens EDA toolchain, covering static analysis, functional simulation, coverage evaluation, design visualization and formal verification.

The study of the AMBA APB Protocol Specification provided a detailed understanding of signal behaviour, transfer types and operating state transitions. The implementation and simulation of the Operating States FSM module, Data Transfer module and Top module to integrate the design together was successful.

This internship has effectively bridged academic learning with industry practice in RTL design and verification.

The future scope of this work aims to include PSLVERR error response logic, PPROT protection signal handling and PSTRB write strobe support as well as development of a complete APB protocol testbench, including directed test cases for all transfer types and error conditions. This would produce a complete RTL implementation of the AMBA APB protocol with all its features.

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ANNEXURES

Annexure 1
PO & PSO Mapping

Note: use a tick mark if you have addressed that PO and PSO in your work

PO No	PO	✓ tick
PO1	**Engineering Knowledge** - Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.	✓
PO2	**Problem analysis -** Identify, formulate, research literature and analyse complex engineering problems reaching sustained conclusions using first principles of mathematics, natural sciences and engineering sciences.	✓
PO3	**Design/Development of Solutions -** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate considerations for the public health and safety and the cultural, societaland the environmental considerations.	✓
PO4	**Conduct investigations of complex problems -** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of the information to provide valid conclusions.	✓
PO5	**Engineering tool usage -** Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling complex engineering activities with an understanding of the limitations.	✓
PO6	**The engineer and the world -** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.	
PO7	**Ethics -** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.	✓
PO8	**Individual and collaborative teamwork -** Function effectively as an individualand as a member or leader in diverse teamsand in multidisciplinary settings.	✓
PO9	**Communication -** Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentationsand give and receive clear instructions.	✓
PO10	**Project management and finance -** Demonstrate knowledge and understanding of the engineering and management principles and apply them to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	
PO11	**Life-long learning -** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	✓

PSO1	Analyse efficient electrical systems and engineering models by applying electrical engineering principles.	✓
PSO2	Design and develop sustainable application-oriented electronic systems.	✓
PSO3	Develop intelligent systems using programming and simulation tools to solve real-world engineering problems.	✓

Expand the mapping with different level and give justifications:

PO and PSO number	Addressed in which chapter	Justification	Level: 0: Not related 1: Low 2: Medium 3: Strong
PO1	2 & 3	Applied engineering principles, Verilog HDL concepts and AMBA APB architecture for RTL design and implementation.	3
PO2	2 & 4	Analysed the APB protocol, verification methodologies and simulation results to validate the RTL design.	3
PO3	3	Designed and implemented the Operating States, Dataflow and Top modules to realise the complete APB protocol.	3
PO4	4	Investigated design correctness through lint analysis, simulation, code coverage and formal verification results.	2
PO5	3 & 4	Used modern EDA tools including Questa Sim, Questa Lint, HDL Designer and Questa Inspect for design and verification.	3
PO6			0

PO7	1	Followed professional ethics by using licensed software tools and accurately reporting the project outcomes.	2
PO8	1 & 5	Worked under the guidance of industry mentors and an academic guide during the internship.	2
PO9	1, 2, 3, 4 & 5	Communicated the methodology, implementation and results through technical documentation and project reporting.	3
PO10			0
PO11	1, 3 & 5	Independently learned Verilog HDL, AMBA APB architecture and Siemens EDA verification tools throughout the internship.	3
PSO1	2 & 3	Applied electrical engineering concepts to analyse and implement the APB communication protocol.	3
PSO2	3 & 5	Designed and verified an application-oriented digital hardware system for low-power peripheral communication.	3
PSO3	3 & 4	Used programming and simulation tools to develop, simulate and verify the RTL implementation of the APB protocol.	3

Annexure 2

LO MAPPING

Note: use a tick mark if you have addressed that LO in your work

PLO No	LO	
C1	Apply knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Some of the knowledge will be at the forefront of the particular subject of study	✓
C2	Analyse complex problems to reach substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles	✓
C3	Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed	✓
C4	Select and evaluate technical literature and other sources of information to address complex problems	✓
C5	Design solutions for complex problems that meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards	✓
C6	Apply an integrated or systems approach to the solution of complex problems	✓
C7	Evaluate the environmental and societal impact of solutions to complex problems and minimise adverse impacts	✓
C8	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct	✓
C9	Use a risk management process to identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity	
C10	Adopt a holistic and proportionate approach to the mitigation of security risks	
C11	Adopt an inclusive approach to engineering practice and recognise the responsibilities, benefits and importance of supporting equality, diversity and inclusion	
C12	Use practical laboratory and workshop skills to investigate complex problems	
C13	Select and apply appropriate materials, equipment, engineering technologies and processes, recognising their limitations	✓
C14	Discuss the role of quality management systems and continuous improvement in the context of complex problems	✓
C15	Apply knowledge of engineering management principles, commercial context, project and change managementand relevant legal matters including intellectual property rights	
C16	Function effectively as an individualand as a member or leader of a team	✓
C17	Communicate effectively on complex engineering matters with technical and non-technical audiences	✓
C18	Plan and record self-learning and development as the foundation for lifelong learning/CPD	✓

Expand the mapping with different levels and give justifications:

IET LO number	Addressed in which chapter	Justification	Level: 0: Not related 1: Low 2: Medium 3: Strong
C1	2 & 3	Applied digital electronics, Verilog HDL and engineering principles to implement the amba APB protocol.	3
C2	2 & 4	Analysed the APB protocol, simulation outputs and verification results to validate the RTL design.	3
C3	3 & 4	Used computational tools including Questa sim, Questa lint, HDL designer and Questa inspect for modelling and verification.	3
C4	2	Reviewed arm protocol specifications, siemens EDA manuals and related technical literature to understand the protocol and verification flow.	3
C5	1 & 3	Designed an RTL solution considering industry standards, low-power communication and verification requirements.	2
C6	3	Integrated the operating states module, dataflow module and top module into a	3

		complete APB system.	
C7	1 & 5	Discussed reduced electronic waste through simulation before hardware implementation and considered environmental impact.	1
C8	1 & 5	Followed professional ethics by using licensed software and reporting results honestly without manipulation.	2
C9			0
C10			0
C11			0
C12			0
C13	3 & 4	Selected and applied appropriate engineering tools and technologies for rtl implementation and verification.	3
C14	4	Evaluated code quality using lint analysis, code coverage and formal verification, demonstrating continuous quality improvement.	2
C15			0
C16	1 & 5	Worked under the guidance of industry mentors and an academic guide throughout the internship.	2
C17	1, 2, 3, 4 & 5	Communicated technical concepts, methodology, results and conclusions	3

		through structured documentation.	
C18	1, 3 & 5	Independently learned Verilog, amba APB architecture and siemens EDA verification tools during the internship.	3

Annexure 3

Project/practice school work classification

Table 1: classification based on project domain classification

Type and Domain	✓ Tick
Product (Hardware/Software)	✓
Simulation	✓
Study	
Application	
Review	
Research	
Domain: Electrical	
Domain: Electronics	✓
Domain: Biomedical	
Domain: Computer science	
Domain: Basic science (math/physics/chemistry)	
Domain: Management	
Domain: Any other (Specify)	

Table 2: classification based on societal consideration

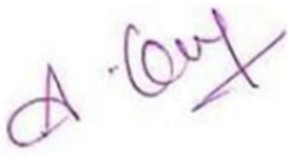
Societal Impact	✓ Tick
ethics	✓
safety	✓
environmental	✓
Healthcare	
commercial	✓
economic	✓
social	

Mention the addressed Sustainable Development Goals (SDG 1-17)

1. SDG 4 : Quality Education : Learning advanced digital design and verification
2. SDG 9 : Industry, Innovation and Infrastructure : Semiconductor design
3. SDG 12: Responsible Consumption and Production : Simulation before fabrication reduces hardware waste

PROJECT DETAILS

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Project Duration	4 months				
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Signature of Student
Date:

Signature of Guide

PLAGIARISM REPORT

Advanced Microcontroller Bus Architecture – Advanced Peripheral Bus

ORIGINALITY REPORT

13%	9%	3%	11%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Manipal Academy of Higher Education (MAHE)	Student Paper
2	Submitted to Manipal University	Student Paper
3	nur.nu.edu.kz	Internet Source
4	acadpubl.eu	Internet Source
5	Submitted to Budapest University of Technology and Economics	Student Paper
6	Submitted to Higher Education Commission Pakistan	Student Paper
7	ldubber.net	Internet Source
8	money.3diamonds.biz	Internet Source
9	Submitted to King's College	Student Paper
10	Submitted to Yonsei University	Student Paper
11	Submitted to National Institute of Technical Teachers' Training & Research, Kolkata	Student Paper
12	dspace.iiti.ac.in	Internet Source
13	odr.chalmers.se	Internet Source
14	ashrithsagar.github.io	Internet Source
15	dokumen.pub	Internet Source

16	f1000research-files.f1000.com Internet Source	
17	github.com Internet Source	
18	uu.diva-portal.org Internet Source	
19	Sourav Das, Sayandeep Sanyal, Aritra Hazra, Pallab Dasgupta. " CoVerPlan: A mprehens ification ning Framework Leveraging PSS Specifications ", ACM Transactions on Design Automation of Electronic Systems, 2022 Publication	

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Exclude bibliography	On		